

Dr. Matthijs Moerkerke

Neuroscientist

Ghent University – Spine, Head and Pain Research Unit, Department of Rehabilitation Sciences
KU Leuven – Center for Developmental Psychiatry, Department of Neurosciences

I am a **postdoctoral researcher** at Ghent University (Belgium) and voluntary research fellow at KU Leuven (Belgium). My research interests include oxytocin, pain science, neuroimaging and neuroendocrine mechanisms.

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Education

PhD in Biomedical Sciences (Neuroscience)

KU Leuven - 2022

The role of oxytocin in autism – a randomized clinical trial.

Dissertation link

Postgraduate Studies in Advanced Medical Imaging

KU Leuven - 2017

Postgraduate specialization in advanced medical imaging.

Master of Biomedical Sciences

KU Leuven - 2016

Biomedical sciences training with a strong research orientation.

Additional Training

- **Research stay – Stanford University, USA (2025)**
  Biobehavioral Pediatric Pain Lab, Prof. Laura Simons
  Supported by Faculty Travel Grant (UGent) and FWO Travel Grant.
- **Research stay – University of Oslo, Norway (2023)**
  Oxytocin research Lab, Prof. Daniel Quintana
  Supported by FWO Travel Grant.
- **PhD in Neuroscience Certificate (2022)**
- **Oxytocin Conference and Workshop – invited talk (2022)**
  Ettore Majorana Foundation, Erice, Italy
  Supported by FWO Travel Grant.
- **Regulatory Medical Writing Training Programme (2021)**
- **FreeSurfer MRI Analysis Course (2019)**
  Martinos Center for Biomedical Imaging, MIT-Harvard, Boston, USA
  Supported by FWO Travel Grant.
- **Good Clinical Practice Certificate (GCP) (2019)**
- **Spanish Language Courses (2017)**
- **MATLAB and Octave Certificate (2017)**
- **Laboratory Animal Science Certificate – FELASA B (2014)**

Teaching & Mentoring

University teaching

- Lecturer for *Neurosciences* (2nd BA Rehabilitation Sciences, Ghent University)
- Lecturer for *Evidence-Based Medicine & Statistics* (3rd BA Rehabilitation Sciences, Ghent University) — responsible for theoretical lectures and coordination of practical seminars.
- Leading curricular reform and supervising eight PhD teaching assistants.

Graduate and undergraduate supervision

- Co-promoter of **2 PhD projects**
- Promoter of **10 MSc theses** (Rehabilitation Sciences, Biomedical Sciences and Medicine)
- Co-promoter of **15 MSc theses**
- Daily supervisor of **24 bachelor and master students**
- Jury member for **16 MSc defenses**
- Jury member for **1 PhD defense**

Mentoring

- Mentor for junior researchers in pain science and stress physiology
- Backup ombudsman for the *Postgraduate Studies in Advanced Medical Imaging* programme at KU Leuven (2017–2023)

Scientific Engagement & Outreach

Scientific collaboration & service

- **Steering-group member**, Pain in Motion (PIM) international research consortium — facilitating cross-university scientific exchange and organizing international networking events (2023–present)
  Pain in Motion website
- **Expert panel member**, Active Monitoring of Oxytocin Research Evidence (AMORE), University of Oslo — contributing to global consensus-building and open-science initiatives (2025–present)
  AMORE website
- **Scientific program & award manager**, Pain Science in Motion Conference 2026 — responsible for program development, international communications, and global participant engagement (2025–present)
  PSIM Conference website
- **Ad hoc grant reviewer**, Foundation for Prader-Willi Research (USA) — oxytocin research grant review (2026)

Scientific outreach in mainstream media

- **EOS Wetenschap** — Oxytocin nasal spray may improve social bonding in children with autism
  EOS article
- **ACAMH Podcast** — Oxytocin administration, neural sensitivity and autism
  ACAMH podcast
- **KU Leuven Blog** — De invloed van oxytocine-neusspray op kinderen met autisme
  KU Leuven blog post
- **Pain in Motion Blog** — The role of oxytocin in chronic pain
  Pain in Motion blog post

Public science events

- UGent Research Day — jury member for poster presentations (2026)
- Day of Science (Dag van de Wetenschap) — public science outreach (yearly neuroscience workshops since 2017)
- Brain Awareness Week — neuroscience outreach events (2017)
- Child University KU Leuven — workshops for elementary school children (2016-2018)
- Autisme Beurs — public engagement on autism research (2017)

Memberships

- International Association for the Study of Pain (IASP)
- Pain Science in Motion (PiM)
- International Society for Autism Research (INSAR)
- Belgian Association for Psychological Sciences (BAPS)
- Leuven Brain Institute (LBI)
- KU Leuven Child & Youth Institute (L-C&Y)
- Leuven Autism Research (LAuRes)
- Alliantieonderzoeksgroep UGent–VUB Motor Mind

Skills & Methods

Languages

●●●●● **Dutch** — C2 (Native)
●●●●● **English** — C1 (Fluent)
●●○○○ **French** — A2 (Basic)
●○○○○ **Spanish** — A1 (Basic)

Medical imaging proficiencies

Scanning, pre-processing and analysis of images taken with **Magnetic Resonance Imaging (MRI)** (functional, structural and diffusion weighted MRI) and **electroencephalography (EEG),** including stress physiology measures (heart rate, respiration rate and skin conductance).

Software experience includes SPM12, Letswave 6, CAT12, CONN, Freesurfer.

Statistical skills

Statistical analysis of behavioural, clinical, and neuroimaging datasets using **RStudio, JAMOVI, SPSS, and MATLAB**, including data preprocessing, visualization, regression analyses, mixed-effects modelling, and hypothesis-driven statistical testing.

Programming skills

Programming scripts for fMRI and behavioural experiments in MATLAB and Presentation (Neurobehavioral Systems)

Laboratory skills

Animal experiments, organ dissection, tissue embedding, microscopic evaluation, and histological analysis.

Clinical skills:

Placing infuses, drawing blood, intramuscular and subdermal injections, ECG and EEG recordings, and CPR training.

Awards & Grants

Fulbright Postdoctoral Research Award 2026
Awarded for an 8 month research stay at Stanford University (CA, USA).
Fulbright website

MARBEL Best Poster Award (3rd place) 2026
Awarded for the poster *Pain and the brain: examining the link between conditioned pain modulation and structural gray matter properties* at the Annual Meeting of the Belgian Magnetic Resonance Imaging Research Community (MaRBEL), Ghent, Belgium.

INSAR PhD Dissertation Award 2024
Awarded by the International Society for Autism Research (INSAR), presented at the Annual Meeting in Melbourne, Australia.
INSAR website

BCNBP "Belgium's Got Talent" PhD Thesis Prize 2023
Awarded by the Belgian College of Neuropsychopharmacology and Biological Psychiatry (BCNBP) for best PhD thesis.
BCNBP website

Thérèse Vandereecken Prize for Clinical Medicine 2023
Royal Academy of Medicine of Belgium recognising research on oxytocin's neural, biological and behavioural effects in children with autism, awared to Prof. Alaerts (PhD copromotor)

KU Leuven Postdoctoral Research Grant 2023
Postdoctoral Mandate (PDM) awarded to continue doctoral research for one year after PhD defence.

Summary of Presentations at Conferences

- **Moerkerke, M.** The role of oxytocin in modulating pain and its interaction with stress in the context of musculoskeletal pain. *EFIC 2025*, Lyon, France. **(oral talk)**
- **Moerkerke, M.** PhD Dissertation Award Presentation: *The role of oxytocin in autism*. *International Society for Autism Research (INSAR) 2024*, Melbourne, Australia. **(oral talk)**
- **Moerkerke, M., Daniels, N., Tibermont, L., Van der Donck, S., Tang, T., Prinsen, J., ... Boets, B.** (2023). Chronic oxytocin administration does not rescue reduced neural sensitivity towards emotional faces in autism. *NeuroCog 2023*, Brussels, Belgium. **(oral talk)**
- **Moerkerke, M., Daniels, N., Tibermont, L., Van der Donck, S., Tang, T., Prinsen, J., Bamps, A., Debbaut, E., Steyaert, J., Alaerts, K., Boets, B.** (2023). Influence of repeated intranasal oxytocin administration on neural activity during face processing in children with autism. *International Society for Autism Research (INSAR)*, Stockholm, Sweden. **(poster presentation)**
- **Moerkerke, M., Tibermont, L., Van der Donck, S., Daniels, N., Tang, T., Prinsen, J., Steyaert, J., Alaerts, K., Boets, B.** (2022). Can longterm intranasal oxytocin administration rescue reduced neural sensitivity towards emotional faces in autism? *Oxytocin and Vasopressin: From Brain Modulation, to Epigenetic Regulation and Clinical Applications*, Erice, Sicily, Italy. **(invited oral talk)**
- **Moerkerke, M., Daniels, N., Chubar, V., Van der Donck, S., Tang, T., Prinsen, J., Claes, S., Steyaert, J., Alaerts, K., Boets, B.** (2022). Oxytocin system alterations in autism? Assessing endogenous oxytocin levels and OXTR epigenetics throughout a long-term intranasal oxytocin treatment. *Oxytocin and Vasopressin: From Brain Modulation, to Epigenetic Regulation and Clinical Applications*, Erice, Sicily, Italy. **(poster presentation)**
- **Moerkerke, M., Chubar, V., Daniels, N., Van der Donck, S., Tang, T., Tibermont, L., Steyaert, J., Alaerts, K., Boets, B.** (2022). Oxytocin receptor epigenetics in autism spectrum disorders. *KU Leuven Child & Youth Institute*, Leuven, Belgium. **(oral talk)**
- **Moerkerke, M., Tibermont, L., Van der Donck, S., Daniels, N., Steyaert, J., Alaerts, K., Boets, B.** (2021). The effect of oxytocin on reduced neural sensitivity towards emotional faces in autism. *Leuven Brain Institute (LBI)*, KU Leuven, Belgium. **(poster presentation)**
- **Moerkerke, M., Daniels, N., Van der Donck, S., Steyaert, J., Alaerts, K., Boets, B.** (2021). Validating an EEG-based biomarker of socio-communicative impairments in ASD. *International Society for Autism Research (INSAR)*, virtual. **(poster presentation)**
- **Daniels, N., Moerkerke, M., Prinsen, J., Boets, B., Steyaert, J., Alaerts, K.** (2021). Measurement of sympathetically driven

autonomic arousal during eye contact in children with and without an autism diagnosis. *International Society for Autism Research (INSAR)*, virtual. **(oral presentation / open access abstract)**

- **Moerkerke, M., Daniels, N., Van der Donck, S., Steyaert, J., Alaerts, K., Boets, B.** (2021). Neurobiological marker and intervention for socio-communicative impairments in autism spectrum disorders. *European Neuropsychopharmacology*, 44(S51–S52). https://doi.org/10.1016/j.euroneuro.2021.01.079 **(poster presentation)**
- **Moerkerke, M., Daniels, N., Stuer-Jespers, A., Steyaert, J., Alaerts, K., Boets, B.** (2018). In search for an innovative neural marker and intervention for socio-communicative difficulties in children with and without autism spectrum disorders. *Understanding the Neuroregulatory Actions of Oxytocin and its Potential Clinical Applications*, Erice, Sicily, Italy. **(poster presentation)**

Publications

PUBLICATIONS	CITATIONS	H-INDEX
40	**498**	**12**

1. Elisabeth Deilhaug, Matthijs Moerkerke, Alina I. Sartorius, Heemin Kang, Emilie Smith-Meyer Kildal, Kjersti M. Walle, Torbjørn Elvsåshagen, Lars T. Westlye, Terje Nærland, Ole A. Andreassen, Daniel S. Quintana (2026). *Using EEG to Measure the Neural Effects of Oxytocin Administration: A Meta-Analysis and Systematic Review*. Psychophysiology. https://doi.org/10.1111/psyp.70334
2. Tiffany Tang, Matthijs Moerkerke, Nicky Daniels, Viktoria Chubar, Bart Vanaudenaerde, Johan Verhaeghe, Stephan Claes, Jean Steyaert, Els Ortibus, Gunnar Naulaers, Kaat Alaerts, Bart Boets (2026). *Assessing peripheral oxytocin and cortisol levels and epigenetic variations of oxytocin receptor and glucocorticoid receptor genes in school-aged preterm-born children*. Progress in Neurobiology. https://doi.org/10.1016/j.pneurobio.2026.102936
3. Elise Cnockaert, Mira Meeus, Barbara Cagnie, Marjolein Chys, Carmel Steverlynck, Matthijs Moerkerke, Jessica Van Oosterwijck (2026). *Human assumed central sensitization in interictal migraine: a systematic review and meta-analysis*. Neurological Sciences. https://doi.org/10.1007/s10072-026-08825-8
4. Ingebjørg A. Iversen, Kaat Alaerts, Marian Bakermans-Kranenburg, Benjamin Becker, Robert James Blair, Jennifer A. Bartz, Jessica J. Connelly, Beate Ditzen, Natalie C. Ebner, Heemin Kang, Elizabeth A. Lawson, Nicole Nadine Lønfeldt, Matthijs Moerkerke, Christian Montag, Anna-Rosa Cecilie Mora-Jensen, Marilyn Horta, Leehe Peled-Avron, Tanya L. Procyshyn, Alina I. Sartorius, Dirk Scheele, Ekaterina Schneider, Constantina Theofanopoulou, Hidenori Yamasue, Daniel S. Quintana (2026). *The active monitoring of oxytocin research evidence (AMORE) platform*. PSYCHONEUROENDOCRINOLOGY. https://doi.org/10.1016/j.psyneuen.2025.107713
5. Sophie Van Oosterwijck, Jaap Wijnen, Michiel Brandt, Amber Billens, Hannes Meirezonne, Evy Dhondt, Lisa Bernaers, Indra De Greef, Ben Jeurissen, Tine Willems, Thomas Matheve, Matthijs Moerkerke, Paul Hodges, Jessica Van Oosterwijck, Lieven Danneels (2026). *Efficacy of Specific Skilled Motor Versus General Exercise Training on Muscle and Neural Alterations in Recurrent Low Back Pain: Protocol of a Randomized Clinical Trial*. Perceptual and Motor Skills. https://doi.org/10.1177/00315125261441406
6. Matthijs Moerkerke, Iris Coppieters, Inge Timmers, Jessica Van Oosterwijck (2026). *Oxytocin in chronic pain: From analgesic to biopsychosocial adjuvant - An opinion paper*. Neuroscience & Biobehavioral Reviews. https://doi.org/10.1016/j.neubiorev.2026.106586
7. Evenepoel Margaux, Tuerlinckx Elise, Derrien Muriel, Moerkerke Matthijs, Prinsen Jellina, Vich Vila Arnau, Steyaert Jean, Daniels Nicky, Boets Bart, Raes Jeroen, Alaerts Kaat (2026). *A pilot study on the role of the oxytocinergic system in gut microbiome composition in children with autism: baseline associations and effects of intranasal oxytocin*. Brain, Behavior, and Immunity. https://doi.org/10.1016/j.bbi.2026.106579
8. Jellina Prinsen, Nicky Daniels, Matthijs Moerkerke, Bart Boets, Kaat Alaerts (2026). *Impact of chronic oxytocin on gaze and pupil dynamics during live dyadic interactions in children with autism*. PSYCHONEUROENDOCRINOLOGY. https://doi.org/10.1016/j.psyneuen.2026.107745
9. Joren Vyverman, Robrecht De Baere, Inge Timmers, Iris Coppieters, Jessica Van Oosterwijck, Matthijs Moerkerke (2026). *The stress–pain connection in chronic primary pain: A systematic review and meta-analysis of physiological stress markers in relation to experimental pain responses*. Neuroscience & Biobehavioral Reviews. https://doi.org/10.1016/j.neubiorev.2026.106604
10. Tiffany Tang, Samantha Piers, Lisa Gistelinck, Nicky Daniels, Matthijs Moerkerke, Jean Steyaert, Els Ortibus, Gunnar Naulaers, Kaat Alaerts, Bart Boets (2026). *Heart rate variability, skin conductance, and gaze behavior during rest and during live eye contact in very prematurely born school-aged children*. Pediatric Research. https://doi.org/10.1038/s41390-025-04613-w
11. Margaux Evenepoel, Nicky Daniels, Matthijs Moerkerke, Jellina Prinsen, Jean Steyaert, Bart Boets, Marie Joossens, Kaat Alaerts (2026). *The role of the oxytocinergic system in oral microbiome composition in children with autism: evidence from a randomized controlled trial of intranasal oxytocin*. Translational Psychiatry. https://doi.org/10.1038/s41398-026-03964-0
12. Matthijs Moerkerke, Joren Vyverman, Robrecht De Baere, Patricia Clement, Tom Smeets, Iris Coppieters, Inge Timmers, Jessica Van Oosterwijck (2025). *A comprehensive mapping of stress system interactions with pain and their contribution to chronification of musculoskeletal pain: Protocol of the STRAIN study*. PLOS One. https://doi.org/10.1371/journal.pone.0324089
13. Matthijs Moerkerke, Iris Coppieters, Inge Timmers, Jessica Van Oosterwijck (2025). *Oxytocin in Chronic Pain: From Analgesic to Biopsychosocial Adjuvant—An Opinion Paper*. preprint. https://doi.org/10.20944/preprints202510.0247.v1
14. Elisabeth Deilhaug, Matthijs Moerkerke, Alina I. Sartorius, Heemin Kang, Emilie Smith-Meyer Kildal, Kjersti M. Walle, Torbjørn Elvsåshagen, Lars T. Westlye, Terje Nærland, Ole A. Andreassen, Daniel S. Quintana (2025). *Using EEG to measure the neural effects of oxytocin administration: A meta-analysis and systematic review*. https://doi.org/10.1101/2025.08.25.25334355
15. Matthijs Moerkerke, Joren Vyverman, Robrecht De Baere, Patricia Clement, Tom Smeets, Iris Coppieters, Inge Timmers, Jessica Van Oosterwijck (2025). *A comprehensive mapping of stress system interactions with pain and their contribution to chronification of musculoskeletal pain: Protocol of the STRAIN study (vol 20, e0324089, 2025)*. PLOS ONE. https://doi.org/10.1371/journal.pone.0331107
16. Tiffany Tang, Matthijs Moerkerke, Nicky Daniels, Bieke Bollen, Jean Steyaert, Kaat Alaerts, Gunnar Naulaers, Els Ortibus, Bart Boets (2025). *Pinpointing the Preterm Behavioural Phenotype in School-Aged Children Without Major Impairments: A Multi-Informant Approach Combining the Perspectives of Child, Parent and Clinician*. JOURNAL OF AUTISM AND DEVELOPMENTAL DISORDERS. https://doi.org/10.1007/s10803-025-07182-3
17. Tiffany Tang, Kasper Pledts, Matthijs Moerkerke, Stephanie Van der Donck, Bieke Bollen, Jean Steyaert, Kaat Alaerts, Els Ortibus, Gunnar Naulaers, Bart Boets (2024). *Face Processing in Prematurely Born Individuals—A Systematic Review*. Brain Sciences. https://doi.org/10.3390/brainsci14121168

18. Kaat Alaerts, Matthijs Moerkerke, Nicky Daniels, Qianqian Zhang, Grazia Ricchiuti, Jean Steyaert, Jellina Prinsen, Bart Boets (2024). *Chronic oxytocin improves neural decoupling at rest in children with autism*. Journal of Child Psychology and Psychiatry. https://doi.org/10.1111/jcpp.13966
19. Margaux Evenepoel, Nicky Daniels, Matthijs Moerkerke, Michiel Van de Vliet, Jellina Prinsen, Elise Tuerlinckx, Jean Steyaert, Bart Boets, Kaat Alaerts, Marie Joossens (2024). *Oral microbiota in autistic children: Diagnosis-related differences and associations with clinical characteristics*. BRAIN BEHAVIOR & IMMUNITY-HEALTH. https://doi.org/10.1016/j.bbih.2024.100801
20. Ward Deferm, Tiffany Tang, Matthijs Moerkerke, Nicky Daniels, Jean Steyaert, Kaat Alaerts, Els Ortibus, Gunnar Naulaers, Bart Boets (2024). *Subtle microstructural alterations in white matter tracts involved in socio-emotional processing after very preterm birth*. NEUROIMAGE-CLINICAL. https://doi.org/10.1016/j.nicl.2024.103580
21. Matthijs Moerkerke, Nicky Daniels, Laura Tibermont, Tiffany Tang, Margaux Evenepoel, Stephanie Van der Donck, Edward Debbaut, Jellina Prinsen, Viktoria Chubar, Stephan Claes, Bart Vanaudenaerde, Lynn Willems, Jean Steyaert, Bart Boets, Kaat Alaerts (2024). *Chronic oxytocin administration stimulates the oxytocinergic system in children with autism*. Nature Communications. https://doi.org/10.1038/s41467-023-44334-4
22. Matthijs Moerkerke, Nicky Daniels, Stephanie Van der Donck, Tiffany Tang, Jellina Prinsen, Elahe' Yargholi, Jean Steyaert, Kaat Alaerts, Bart Boets (2024). *Impact of chronic intranasal oxytocin administration on face expression processing in autistic children: a randomized controlled trial using fMRI*. MOLECULAR AUTISM. https://doi.org/10.1186/s13229-024-00635-z
23. Matthijs Moerkerke, Nicky Daniels, Stephanie Van der Donck, Laura Tibermont, Tiffany Tang, Edward Debbaut, Annelies Bamps, Jellina Prinsen, Jean Steyaert, Kaat Alaerts, Bart Boets (2023). *Can repeated intranasal oxytocin administration affect reduced neural sensitivity towards expressive faces in autism? A randomized controlled trial*. Journal of Child Psychology and Psychiatry. https://doi.org/10.1111/jcpp.13850
24. Margaux Evenepoel, Matthijs Moerkerke, Nicky Daniels, Viktoria Chubar, Stephan Claes, Jonathan Turner, Bart Vanaudenaerde, Lynn Willems, Johan Verhaeghe, Jellina Prinsen, Jean Steyaert, Bart Boets, Kaat Alaerts (2023). *Endogenous oxytocin levels in children with autism: Associations with cortisol levels and oxytocin receptor gene methylation*. Translational Psychiatry. https://doi.org/10.1038/s41398-023-02524-0
25. Matthijs Moerkerke, Nicky Daniels, Laura Tibermont, Tiffany Tang, Margaux Evenepoel, Stephanie Van der Donck, Edward Debbaut, Jellina Prinsen, Viktoria Chubar, Stephan Claes, Bart Vanaudenaerde, Lynn Willems, Jean Steyaert, Bart Boets, Kaat Alaerts (2023). *Chronic oxytocin administration stimulates the endogenous oxytocin system: an RCT in autistic children*. Preprint. https://doi.org/10.1101/2023.06.06.23291017
26. Kaat Alaerts, Nicky Daniels, Matthijs Moerkerke, Margaux Evenepoel, Tiffany Tang, Stephanie Van der Donck, Viktoria Chubar, Stephan Claes, Jean Steyaert, Bart Boets, Jellina Prinsen (2023). *At the head and heart of oxytocin's stress-regulatory neural and cardiac effects: a chronic administration RCT in children with autism*. Preprint. https://doi.org/10.1101/2023.04.04.23288109
27. Nicky Daniels, Matthijs Moerkerke, Jean Steyaert, Annelies Bamps, Edward Debbaut, Jellina Prinsen, Tiffany Tang, Stephanie Van der Donck, Bart Boets, Kaat Alaerts (2023). *Effects of multiple-dose intranasal oxytocin administration on social responsiveness in children with autism*. Molecular Autism. https://doi.org/10.1186/s13229-023-00546-5
28. Zhiling Qiao, Stephanie Van der Donck, Matthijs Moerkerke, Tereza Dlhosova, Sofie Vettori, Milena Dzhelyova, Ruud van Winkel, Kaat Alaerts, Bart Boets (2022). *Frequency-Tagging EEG of Superimposed Social and Non-Social Visual Stimulation Streams Provides No Support for Social Salience Enhancement after Intranasal Oxytocin Administration*. Brain Sciences. https://doi.org/10.3390/brainsci12091224
29. Nicky Daniels, Matthijs Moerkerke, Jean Steyaert, Annelies Bamps, Edward Debbaut, Jellina Prinsen, Tiffany Tang, Stephanie Van der Donck, Bart Boets, Kaat Alaerts (2022). *Effects of multiple-dose intranasal oxytocin treatment on social responsiveness in children with autism: A randomized, placebo-controlled trial*. Preprint. https://doi.org/10.1101/2022.04.20.22274106
30. Stephanie Van der Donck, Matthijs Moerkerke, Tereza Dlhosova, Sofie Vettori, Milena Dzhelyova, Kaat Alaerts, Bart Boets (2022). *Monitoring the effect of oxytocin on the neural sensitivity to emotional faces via frequency-tagging EEG: A double-blind, cross-over study*. Psychophysiology. https://doi.org/10.1111/psyp.14026
31. Matthijs Moerkerke (2022). *PhD Thesis: The Role of Oxytocin in Autism*. PhD Dissertation, KU Leuven.
32. Matthijs Moerkerke, Mathieu Peeters, Lyssa de Vries, Nicky Daniels, Jean Steyaert, Kaat Alaerts, Bart Boets (2021). *Endogenous Oxytocin Levels in Autism—A Meta-Analysis*. Brain Sciences. https://doi.org/10.3390/brainsci11111545
33. Jeroen Van Dessel, Edmund Sonuga-Barke, Matthijs Moerkerke, Saskia Van der Oord, Sarah Morsink, Jurgen Lemiere, Marina Danckaerts (2021). *The limits of motivational influence in ADHD: no evidence for an altered reaction to negative reinforcement*. Social Cognitive and Affective Neuroscience. https://doi.org/10.1093/scan/nsab111
34. Matthijs Moerkerke, Marie-Laure Bonte, Nicky Daniels, Viktoria Chubar, Kaat Alaerts, Jean Steyaert, Bart Boets (2021). *Oxytocin receptor gene (OXTR) DNA methylation is associated with autism and related social traits – A systematic review*. Research in Autism Spectrum Disorders. https://doi.org/10.1016/j.rasd.2021.101785
35. Jeroen Van Dessel, Marina Danckaerts, Matthijs Moerkerke, Saskia Van der Oord, Sarah Morsink, Jurgen Lemiere, Edmund Sonuga-Barke (2021). *Dissociating brain systems that respond to contingency and valence during monetary loss avoidance in adolescence*. Brain and Cognition. https://doi.org/10.1016/j.bandc.2021.105723
36. Matthijs Moerkerke, Nicky Daniels, Stephanie Van der Donck, Jean Steyaert, Kaat Alaerts, Bart Boets (2021). *P.318 Neurobiological marker and intervention for socio-communicative impairments in autism spectrum disorders*. European Neuropsychopharmacology. https://doi.org/10.1016/j.euroneuro.2021.01.079
37. Jeroen Van Dessel, Edmund Sonuga-Barke, Matthijs Moerkerke, Saskia Van der Oord, Jurgen Lemiere, Sarah Morsink, Marina Danckaerts (2019). *The amygdala in adolescents with attention-deficit/hyperactivity disorder: Structural and functional correlates of delay aversion*. World Journal of Biological Psychiatry. https://doi.org/10.1080/15622975.2019.1585946
38. Jeroen Van Dessel, Sarah Morsink, Saskia Van der Oord, Jurgen Lemiere, Matthijs Moerkerke, Margaux Grandelis, Edmund Sonuga-Barke, Marina Danckaerts (2019). *Waiting impulsivity: a distinctive feature of ADHD neuropsychology?*. Child Neuropsychology.

39. Jeroen Van Dessel, Matthijs Moerkerke, Edmund Sonuga-Barke, Saskia Van der Oord, Jurgen Lemiere, Marina Danckaerts (2018). *Distinctive neural response towards certain and conditional monetary loss in adolescents with attention-deficit/hyperactivity disorder*. European Neuropsychopharmacology. https://doi.org/10.1016/j.euroneuro.2017.12.108

40. Jeroen Van Dessel, Matthijs Moerkerke, Edmund Sonuga-Barke, Jurgen Lemiere, Saskia Van der Oord, Marina Danckaerts (2017). *Understanding the role of the amygdala in attention-deficit/hyperactivity disorder: association between brain structure, function and delay aversion*. European Neuropsychopharmacology. https://doi.org/10.1016/s0924-977x(17)31898-9